

Documents allowed. Calculators allowed - Duration: 2h00

The points are given as an indication

The subject is divided into 6 independent exercises. The quality of the writing, the clarity and the precision of the reasoning will play an important part in the evaluation.

Exercise 1 (Floating-point arithmetic (4 points)). Let $a = 47806$ and $b = 0.00244140625 = 2^{-9} + 2^{-11}$. Let $c = a \oplus b$ the floating-point number obtained by addition of a and b in single-precision with rounding to nearest. Give the single-precision representation of c .

Exercise 2 (CADNA 1 (5 points)). We recall below the characteristics of some IEEE 754 formats

format	mantissa length (hidden bit included)	unit round-off
binary32	24 bits	$\approx 6.E-8$
binary64	53 bits	$\approx 1.E-16$
binary128	113 bits	$\approx 1.E-34$

A dot product is computed in double precision (IEEE binary64).

Its correctly rounded value is:

$$D_{exact} = -7.6615611072031486e-01$$

Its value computed with rounding to nearest is:

$$D_{RTN} = -7.6615610946693546e-01$$

The digits estimated correct by stochastic arithmetic are:

$$D_{CADNA} = -0.7661561E+000.$$

1. Are these results consistent?
2. One numerical instability has been detected by stochastic arithmetic:
1 LOSS(ES) OF ACCURACY DUE TO CANCELLATION(S)
Describe this kind of instability.
May it invalidate accuracy estimation by stochastic arithmetic? Justify your answer.
3. The condition number of this dot product is estimated $2.15e+08$. Is the accuracy estimation of the dot product by stochastic arithmetic consistent with the condition number?
4. How many correct digits do we expect:
in single precision (IEEE binary32)?
in quadruple precision (IEEE binary128)?

Exercise 3 (CADNA 2 (3 points)). Newton iterations are used to compute polynomial roots.

The associated sequence is:

$$x_{n+1} = x_n - P(x_n)/P'(x_n)$$

A possible C implementation is:

```

x=x0;
do {
    y=x;
    x=y-P(y)/P_deriv(y);
}
while (fabs(x-y)>=epsilon);

```

1. Explain why using this code with stochastic arithmetic may result in an infinite loop.
2. Which modification would you suggest for the stochastic arithmetic version of the code?
Let us denote by R the result obtained.
3. It has been proved that, in the convergence zone of Newton iterations, the digits common to two successive iterates are also in common with the exact root.
What is the impact of this property on R ?

Exercise 4 (Fixed-point arithmetic (6 points)).

1. Consider the signed fixed-point format $(6, -4)$ (most significant bit position: 6, least significant bit position: -4). Indicate the number of bits in this format, and the smallest and largest values. What is the quantization step?
2. Represent the following reals in 8-bit signed fixed point, indicating the fixed point format (position of the most significant bit and the least significant bit).

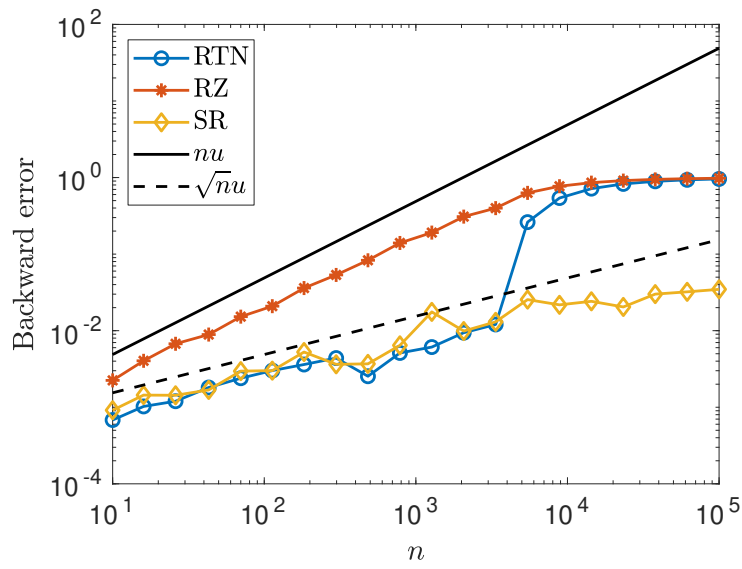
$$9.3452 \quad 0.03434 \quad -2.18625$$

3. Indicate the relative and absolute error for each. In a general way, how to bound these two errors as a function of the width w used to represent the number.
4. We want to calculate

$$9.3452x + 0.03434y - 2.18625z$$

where x , y and z are variables known to belong to the intervals $[-24; 3.4]$, $[-150; 2500]$ and $[-112; 18]$ respectively. Propose a fixed-point format for these three variables, as well as a fixed-point realization (set of mantissa operations) to best perform this operation, assuming we have an 8-bit by 8-bit multiplier giving a 16-bit result, and a 16-bit accumulator (plus 4 guard bits).

Exercise 5 (Rounding error accumulation (2 points)). Analyze and discuss the figure below, which compares the backward error of the matrix-vector product $y = Ax$ computed in fp16 arithmetic, where the coefficients of $A \in \mathbb{R}^{n \times n}$ and x have been randomly generated in $[0, 1]$, for three different rounding modes: round-to-nearest (RTN), round-to-zero (RZ), and stochastic rounding (SR).



Exercise 6 (Mixed precision blocked summation (4 points)). We consider the summation algorithm below, which is parameterized by three precisions u_1 , u_2 , and u_3 , and by two block sizes b_1 and b_2 (we assume for simplicity that n is a multiple of $b_1 b_2$).

Input: $x_1, \dots, x_n$
Output: $s = \sum_{i=1}^n x_i$
for $i = 1$ **to** $\frac{n}{b_1}$ **do**
 $y_i = \sum_{j=(i-1)b_1+1}^{ib_1} x_j$ in precision u_1
end for
for $i = 1$ **to** $\frac{n}{b_1 b_2}$ **do**
 $z_i = \sum_{j=(i-1)b_2+1}^{ib_2} y_j$ in precision u_2
end for
 $s = \sum_{j=1}^{\frac{n}{b_1 b_2}} z_j$ in precision u_3

1. Draw the summation tree corresponding to this algorithm for $n = 12$, $b_1 = 3$, and $b_2 = 2$ (as a reminder, in a summation tree the leaves correspond to the summands x_i , the root corresponds to the sum s , and the rest of the nodes correspond to partial sums). You will assign to non-leaf nodes the labels 1, 2, or 3 to indicate which precision u_1 , u_2 , or u_3 is used.
2. The computed sum $\hat{s}$ satisfies a backward error bound of the form

$$\hat{s} = \sum_{i=1}^n x_i(1 + \theta_i), \quad |\theta_i| \leq f(n, b_1, b_2, u_1, u_2, u_3).$$

What is the expression of $f(n, b_1, b_2, u_1, u_2, u_3)$?

3. If $u_1 = u_2 = u_3$, which choice of b_1 and b_2 minimizes the bound?
4. If we want to use three different arithmetics (such as fp64, fp32, and fp16), which arithmetic do you propose to assign to which parameter u_1 , u_2 , u_3 ? Why?